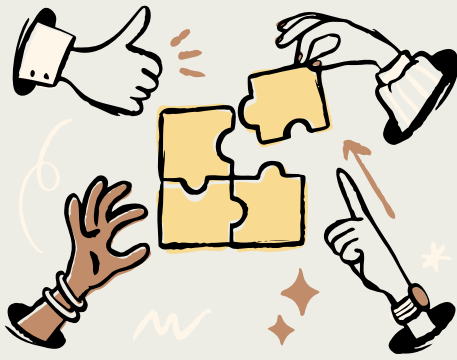




Design documentation

Problem Statement:

People struggle to take passport-approved photos because of strict rules for pose, lighting, and background. This often leads to rejected photos, wasted time, and extra costs. An AI-guided mobile app is needed to help users take compliant passport photos easily and correctly on their own.

Project summery:

"Hi, I'm Krishaang and this is **Shotify**.

Music wouldn't be the same without **Spotify**, and e-shopping wouldn't exist without **Shopify**. Now, it's time for **Shotify** — an app that makes taking passport photos simple.

Using **AI and voice guidance**, Shotify tells you how to pose, fixes your background, and even takes the photo for you — no clicks needed. It checks everything with **official government standards**, so your photo's ready to go in seconds.

Shotify — the smart way to take your perfect shot."

Overview

Product	According to the survey the app had received a 4.5 out of 5 so this proves that the app is unique and will be used by many people.
Specifications	● People Said that the photo session was easy to run about 85% ● The layout did not have a sever lag ● The app was really beneficial and would be shared to others
Key features and benefits	● AI Guidance & Government Verification ● Easy to Use & Simple Navigation ● Convenience & Accessibility
Price point	Shotify is designed to be easy, smart, and affordable for everyone. The first two photos are completely free , so users can try the app with no cost. After that, it's just \$1 per month for unlimited verified, high-quality passport photos — a fraction of what traditional photo studios charge.

Risk-mitigation:

-The users might not trust the use of government verify checkers.

Solution- The app is now not going to divert the user to some-other website it is going to have an inbuilt checker for the picture.

Customer feedback

Sources

List channels through which feedback is collected.

- Online surveys (pro's)
 - "Easy to use and simple to understand."
 - "Clean, sleek, and professional design."
 - "Very accurate and convenient — saves time and money."
- Online surveys (con's)
 - "Could use more features and customization."
 - "Some buttons and loading times need improvement."
 - "Would be better as a mobile app instead of a website."

Analysis

Interpret feedback as a team by identifying major themes and providing insights and interpretations.

Comment

- "Could use more features and customization."

Solution:

- A background option of your choice
- An auto navigating feature where you don't press any-buttons during your photo section

Comment

- "Some buttons and loading times need improvement."

Solution:

- We can add a speed monitoring system the makes changes according to the speed of the internet.
- I can add a reload button when a lag is occurred.

Comment

- "Would be better as a mobile app instead of a website."

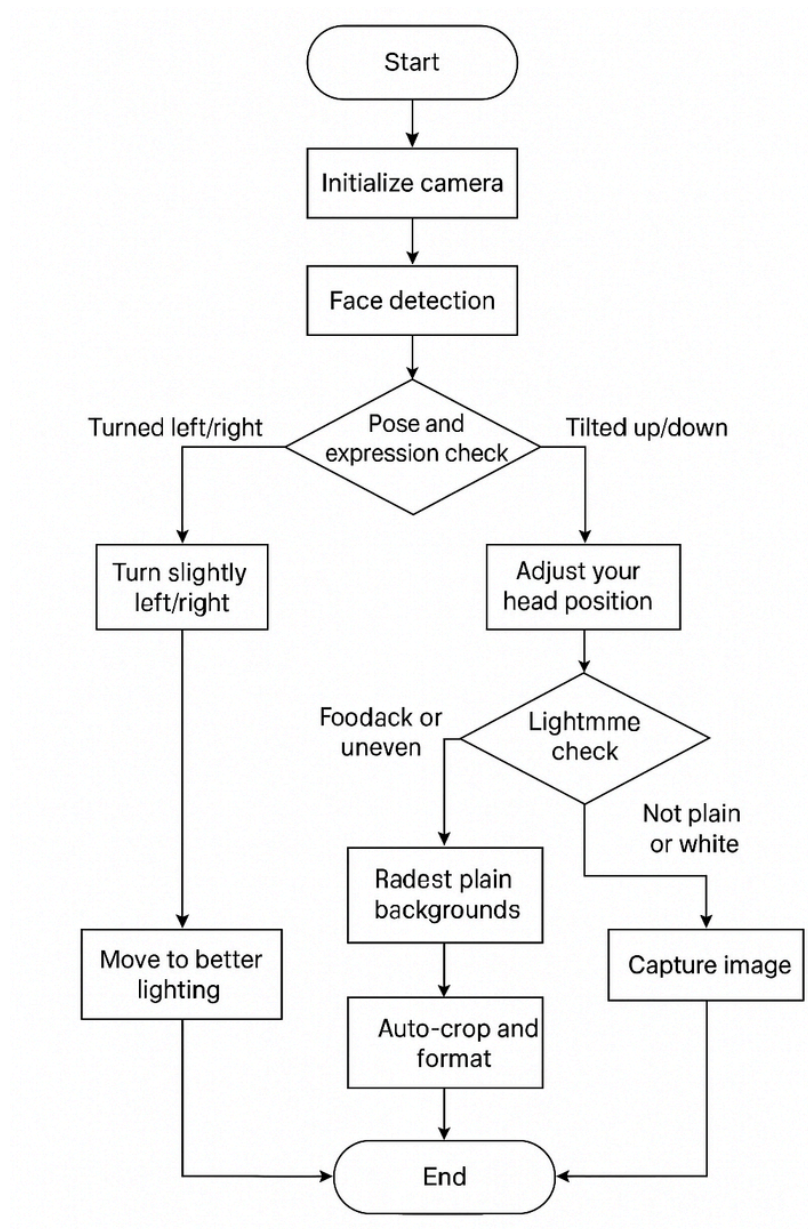
Solution:

- We can make a app version for this website
- It will be available in both options.

Algorithm

Action	explanation
Step 1: Start	Launch the app and open the "Start Photo Session" screen.
Step 2: Initialize Camera	Display a face outline or oval guide on screen.
Step 3: Face Detection	Locate the user's face in real time. Detect key facial points (eyes, nose, mouth, chin).
Step 4: Pose and Expression Check	Calculate face angle using eye and nose coordinates.
Step 5: Lighting Check	If lighting is too dark or uneven, show "Move to better lighting."
Step 6: Background Check	If background is not plain or white, show "Use a plain light background."
Step 7: Capture Image	When all checks pass (pose, light, background, expression):
Step 8: Auto-Crop and Format	Crop image to standard passport dimensions (e.g., 2x2 inch ratio).
Step 9: End	Save or allow user to retake.

Flow-chart



Time-line

Month	Milestone	Description
Month 1 – Planning & Research	Project Scope Defined	Outline the app idea, define user needs, study passport photo standards, and decide core features (AI guidance, auto-capture, background removal).
Month 2 – Design	UI/UX Wireframes Completed	Create home screen, photo session screen, help/support screens, and interactive prototype for testing flow.
Month 3 – Core AI Module	Face Detection & Angle Tracking Ready	Integrate Mediapipe/FaceMesh for real-time head angle detection, facial landmark tracking, and pose feedback.
Month 4 – Auto-Capture & Voice Guidance	AI Voice Assistant Implemented	Develop text-to-speech guidance; set auto-capture triggers when face alignment and lighting are correct.
Month 5 – Background & Cropping	Background Removal & Formatting	Implement white background replacement, automatic cropping, resizing to passport standards, and preview/save feature.
Month 6 – App Integration	Prototype Integration Complete	Combine camera, AI guidance, auto-capture, background processing, and UI into a working prototype.
Month 7 – Testing & Debugging	Internal QA & Bug Fixing	Test all features, refine AI instructions, correct edge cases (bad lighting, different angles, multiple devices).
Month 8 – Beta Testing	User Feedback Collected	Share prototype with small group of users; gather feedback on usability, clarity, speed, and satisfaction.

Month 9 – Iteration	Enhancements Implemented	Improve UI/UX, optimize AI guidance, adjust thresholds for auto-capture, fix reported bugs from beta.
Month 10 – Launch Preparation	Final App & Presentation Ready	Polish prototype for final presentation or limited release, prepare documentation, marketing plan, and proposal visuals.